

Module 2: Lesson 2

Convolutional Neural Networks: Candlesticks and convolutions



Outline

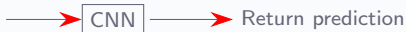
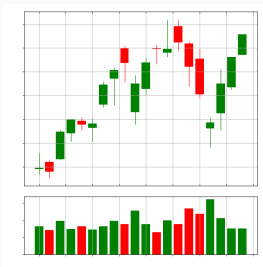
- ▶ Candlestick charts
- ▶ Convolutions for (actual) images

ConvNets for return prediction

In Lesson 1, we described the convolution operator from a fully-connected layer that is constrained to have a limited set of constant weights across hidden representations.

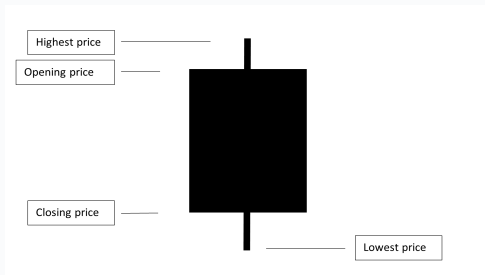
We are going to illustrate the implementation of ConvNets in the context of a problem to predict future stock returns using images.

Our inputs will be the evolution of past returns as depicted in candlestick plots (or charts).

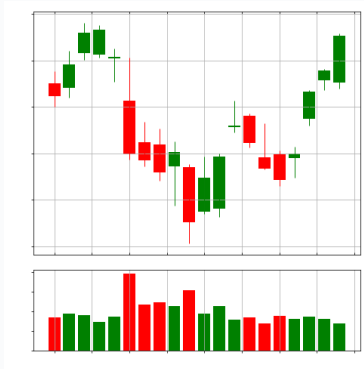


Candlesticks

A candlestick is a type of price chart used in technical analysis that displays the high, low, open, and closing prices of a security for a specific period, usually a trading day.



Processing images in Python

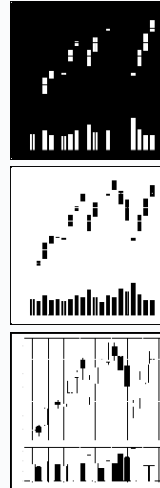
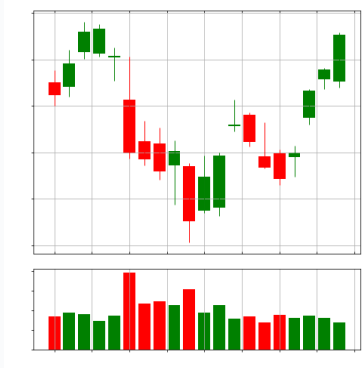


Convolutions from scratch in Python

```
input = np.array([[1, 1, 1], [0, 1, 0], [1, 0, 1]])
kern = np.array([[1, 2], [0, 1]])
convolution = np.zeros((input.shape[0]-kern.shape[0]+1, input.shape[1]-
kern.shape[1]+1))
for i in range(input.shape[0]):
    if i + kern.shape[0] > input.shape[0]:
        break
    for j in range(input.shape[1]):
        if j + kern.shape[1] > input.shape[1]:
            break
        else:
            convolution[i,j] = np.sum(kern * input[i:i + kern.shape[0],j:j + kern.shape[1]])
convolution
```

$$\begin{array}{|c|c|c|} \hline \text{INPUT} & & \\ \hline 1 & 1 & 1 \\ \hline 0 & 1 & 0 \\ \hline 1 & 0 & 1 \\ \hline \end{array} * \begin{array}{|c|c|} \hline \text{KERNEL} & \\ \hline 1 & 2 \\ \hline 0 & 1 \\ \hline \end{array} = \begin{array}{|c|c|} \hline \text{OUTPUT} & \\ \hline 4 & 3 \\ \hline 2 & 2 \\ \hline \end{array}$$

Convolution of a candlestick plot



Summary of Lesson 2

In Lesson 2, we have:

- ▶ Described our stock return prediction problem using ConvNets and candlestick charts
- ▶ Implemented the convolution operator from scratch in Python

⇒ **References for this lesson:**

Chapter 7 in [Zhang, Aston, et al. "Dive into deep learning." \(2022\).](#)

TO DO NEXT: Please, go to the Jupyter Notebook associated with Lesson 2.

In the next lesson, we will describe the convolution operator with multiple input and output channels, as well as additional types of layers that we can include in CNNs to enhance their performance.